

Dear group, my name is Eduardo Cruz and I'm a computer/electronics reverse engineering hobbyist based in Spain.

I'm always up for a good challenge and Kryptos K4 caught my attention about 2 years ago, since then I have dedicated on and off time in an attempt to understand it. In this email I disclose a plausible cipher solution to the plain-text K4 block EASTNORTHEAST.

As this is my first email to the group, I'm not sure if the formatting of my notes will be preserved, for such reason I have uploaded a pdf version of my notes to group files, you will find it under the folder "Eduardo Cruz".

During my analysis of Kryptos K4, I've come to the strong conclusion that none of the methods or combination of methods previously used in K1, K2 and K3 are at play in K4. There's a few reasons as to why I think this is the case, the analysis and discoveries presented further below demonstrate that no vigenere table is used in Kryptos as seen in K1-K2, and also no usage of transposition techniques as seen in K3.

Instead, and as already pointed out by the creators of Kryptos publicly, K4 seems to consist of two cipher methods stacked together working in combination, a first layer with the purpose of masking the english language to complicate analysis, and a second final cipher layer seating on top.

The biggest trace as to how K4 is ciphered comes from padding K4 to 98 letters (97+1), this arrangement offers the possibility of aligning K4 in rows using mathematical divisors 2, 7, 14 and 49. Out of these, aligning K4 to a width of 7 columns and 14 rows produces a very distinct feel with as many as 14 observable repetition patterns, both occurring horizontally and vertically. To top it up, in such arrangement both pieces of known plain text happen to start precisely at the beginning of a block. The resulting 7x14 block structure looks as follows:

	Pos. 1234567	Horizontal patterns 1234567	Vertical patterns 1234567
Block			
1	OBKRUOX		0
2	OGHULBS		0 B
3	OLIFBBW	BB	0L B
4	FLRVQQP	QQ	L

5	RNGKSS0	SS	S
6	TWTQ SJQ		S
7	SSEKZZW	SS ZZ	K
8	ATJKLUD		K
9	IAWINFB		
10	NYPVTTM	TT	P
11	ZFPKWGD		P D
12	KZXTJCD		D
13	IGKUHUA		K
14	UEKCAR		K

When looking at letter count weight by translating letters in the alphabet (A-Z) into numbers (1-26), we can observe the following cipher-text (CT) vs plain-text (PT) distribution in each block.

Block #	1	2	3	4	5	6	7	8	9	10	11	12	13	14
CT weight	108	84	69	108	103	126	129	79	64	130	93	98	78	59
CT weight avg.	15.4	12.0	9.9	15.4	14.7	18.0	18.4	11.3	9.1	18.6	13.3	14.0	11.1	9.8
CT weight adjusted *					88 * (6)						59 * (4)			
PT weight				92	73 * (6)					63	41 * (4)			
PT weight avg.				13.4	12.6					9	10.25			
CT PT distance				22	21					57	34			
CT PT distance avg.				3.14	3.50					8.14	8.50			

* Available plain-text is less than cipher-text its 7 letter counterpart block. Cipher-text weight count is adjusted to the corresponding number of known plain-text letters.

Focusing the analysis on blocks containing known plain-text (blocks #4 #5 #10 & #11), average cipher-text to plain-text distance reveals some very interesting and important information.

On one hand we can see the cipher process being additive as it generally tends to push the plain-text alphabet up. On blocks #4 and #5 this push seems to leave things very close to shore, on average it pushes plain-text up by just 3.32 alphabet positions.

On the other hand, the fact that known plaint-text block pairs share close plain-text to cipher-text

distances, potentially indicates the same or a very similar ciphering mechanism is used on each pair. This hypothesis seems to be true at least for blocks #4 and #5 as we will see in a moment.

When comparing blocks #4 & #5 vs #10 & #11, the distance factor goes up significantly to an average of 8.32 characters (blocks #10 & #11) vs 3.32 (blocks #4 & #5). This seems to indicate that the cipher process and/or its parameters are not always the same and get changed at a certain interval.

Let's now focus solely on blocks #4 and #5, the longest and most complete K4 section of known plain-text.

K4 7-letter block:	#4	#5
Letter position:	1234567	1234567
Cipher-text:	FLRVQQP	RNGKSS0
Known Plain-text:	EASTNOR	THEAST

(the last letter of block #5 plain-text is unknown)

When looking at these two blocks, a few obvious traits can be observed:

- The "S" on the second "EAST" plain-text word matches its cipher-text counterpart.
- "FLRV" and "GKSS" cipher-text both representing plain-text word "EAST" are at very close distance, mostly shift by 1.
- Block #4 plain-text "NO" and block #5 "ST", two alphabet consecutive plain-text letter pairs that sharing the same block location as characters 5 & 6, both produce a similar cipher-text repetition pattern: "QQ" and "SS".
- Average cipher-text vs plain-text distance is very close on both blocks: 3.14 and 3.50 alphabet positions.

The careful consideration of these traits offers enough hints to the presence of a repeating cipher system, perhaps with interferences from at least one additional cipher system. The main challenge to solving K4 is being able to untangle the different cipher techniques into something arguably logical and repeatable.

After extensive analysis of Kryptos blocks #4 and #5, the following findings were be made:

Step #1: We assume block #5 letter at position #5 "S" is not shifted at all. Plain-text and cipher-text

are the same letter "S".

K4 Block:	#4							#5						
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	0
Shift:												0		
Shifted-text:												S		

Step #2: Next to it, letter "T" is consecutive to letter "S" in the alphabet, so shift by 1.

K4 Block:	#4							#5						
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	0
Shift:												0	1	
Shifted-text:												S	T	

Step #3: We use the previous two shifts learned in block #5 on block #4 on the same positions. We don't obtain direct plain text but both resulting letters "Q" and "R" share an interesting equal distance of 3 alphabet positions vs plaintext letters "N" and "0" (from plain-text word NORTH).

K4 Block:	#4							#5						
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	0
Shift:					0	1						0	1	
Shifted-text:					Q	R						S	T	

Step #4: Block #4 expected known plaintext also contains the letters "ST" (word EAST), so we calculate their distance to cipher text.

K4 Block:	#4							#5						
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	0
Shift:			1	-2	0	1						0	1	
Shifted-text:			S	T	Q	R						S	T	

Step #5: We use the shifts learned from block #4 letter locations 3 & 4 on block #5. The resulting "H" is also at a distance of 3 from plaintext "E", letter "I" is at a distance of 8 from plaintext "A" (word EAST).

K4 Block:	#4							#5						
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	0
Shift:	2	-3	1	-2	0	1				1	-2	0	1	
Shifted-text:			S	T	Q	R				H	I	S	T	

Step #6: As we also have plaintext on block #4 letters "EA" (word EAST), we calculate their distance represented as intermediate cipher letters "H" and "I" as found out in the previous step.

K4 Block:	#4							#5						
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	0
Shift:	2	-3	1	-2	0	1				1	-2	0	1	
Shifted-text:	H	I	S	T	Q	R				H	I	S	T	

Step #7: Learned shifts from step 6 block #4 are also applied to the same locations of block #5. In this step we successfully obtain another piece of plaintext at block #5 location #1. letter "T".

K4 Block:	#4							#5						
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	"
Cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	0
Shift:	2	-3	1	-2	0	1		2	-3	1	-2	0	1	
Shifted-text:	H	I	S	T	Q	R		T	K	H	I	S	T	

Step #8: We leave position #7 on both blocks as a question mark. This is because we don't have any other known plaintext "R" to help us solve block #4 letter position 7, and block #5 letter 7 plaintext is unknown.

K4 Block:	#4							#5						
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	0

Shift:	2	-3	1	-2	0	1	?	2	-3	1	-2	0	1	?
Shifted-text:	H	I	S	T	Q	R	?	T	K	H	I	S	T	?

Overall the resulting shifted-text shows two obvious repetition patterns of the word "HIST" matching known plaintext locations for the word "EAST". Letters "S" and "T" are always seen in plaintext and happen to be at the right places.

Step #9: We now try breaking our findings into two different cipher system layers. Since we already have some direct plaintext, letters "S" and "T", we assume all their occurrences are only affected by one of the cipher systems layers.

K4 Block:	#4							#5						
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	0
Shift #1:	2	-3	1	-2	0	1	?	2	-3	1	-2	0	1	?
Shifted-text #1:	H	I	S	T	Q	R	?	T	K	H	I	S	T	?
Shift #2:				0	0							0	0	
Shifted-text #2:				S	T							S	T	

Step #10: All remaining letters but one are at a distance of 3 letters in the alphabet vs the expected plaintext. H = E, K = H, Q = N, R = 0.

K4 Block:	#4							#5						
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	0
Shift #1:	2	-3	1	-2	0	1	?	2	-3	1	-2	0	1	?
Shifted-text #1:	H	I	S	T	Q	R	?	T	K	H	I	S	T	?
Shift #2:	-3			0	0	-3	-3	0	-3	-3		0	0	
Shifted-text #2:	E			S	T	N	0	T	H	E		S	T	

Step #11: Lastly, both remaining occurrences of plaintext "A" are at a distance of 8 vs cipher-text "I".

K4 Block:	#4							#5						
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	0

Shift #1:	2	-3	1	-2	0	1	?	2	-3	1	-2	0	1	?
Shifted-text #1:	H	I	S	T	Q	R	?	T	K	H	I	S	T	?
Shift #2:	-3	-8	0	0	-3	-3		0	-3	-3	-8	0	0	
Shifted-text #2:	E	A	S	T	N	0		T	H	E	A	S	T	

The above findings can be summarized as follows: Cipher layer #1 employs a letter dependent mono-alphabetic substitution, while cipher layer #2 employs a letter position dependent shift.

K4 Block:	#4		#5											
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Starting plain-text:	E	A	S	T	N	0	R	T	H	E	A	S	T	?
Layer 1 substitution:	3	8	0	0	3	3	?	0	3	3	8	0	0	?
Resulting text:	H	I	S	T	Q	R	?	T	K	H	I	S	T	?
Layer 2 shift:	-2	3	-1	2	0	-1	?	-2	3	-1	2	0	-1	?
Resulting cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	0

Cipher Layer 1 mono-alphabetic substitution known parameters so far:

Alphabet PT:	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Shift:	8				3			3						3	3				0	0						
Alphabet CT:	I				H			K						Q	R				S	T						

Cipher Layer 2 shift known parameters so far:

Letter position:	1	2	3	4	5	6	7
Numerical shift:	-2	3	-1	2	0	-1	?

Regarding the missing two letters at position #7, most likely Layer-2 shifts are 0, 1, 2 or 3. Shifts -1 -2 -3 -4 -5 or 4 are not possible due to collisions with other known CT-PT letter pairs.

Shift by 1 is my favorite because its overall roundness:

-The number is not used before in the shift series.

-It creates an incremental sequence at the odd positions (-2, -1, 0, 1).

-In the substitution alphabet, shifting block #4 letter at position #7 by 1 gives letter "0", a -3 shift position from known plaintext "R", creating a correspondence with the known "0 = R" seen in #4 letter

position #6.

-Same distance mirror happens with #5 letter position #7, shifting by 1 gives letter "N" also at -3 from alphabet letter "Q", also a correspondence with "N = Q" seen in block #4 letter position #5.

-If the last letter of block #5 were "Q", the resulting word could be QUADRANT or QUARTER, terminology compatible with compass directions and mapping, a great fit with the preceding sentence "EAST NORTHEAST".

Analysis updated with Block #4 & #5 letter #7 assumptions:

K4 Block:	#4							#5						
Letter position:	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Starting plain-text:	E	A	S	T	N	O	R	T	H	E	A	S	T	Q
Layer 1 mono subst.:	3	8	0	0	3	3	-3	0	3	3	8	0	0	-3
Resulting text:	H	I	S	T	Q	R	O	T	K	H	I	S	T	N
Layer 2 shift:	-2	3	-1	2	0	-1	1	-2	3	-1	2	0	-1	1
Resulting cipher-text:	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O

Cipher Layer 2 shift known parameters:

Letter position:	1	2	3	4	5	6	7
Numerical shift:	-2	3	-1	2	0	-1	1

Block #	1	2	3	4	5	6	7	8	9	10	11	12	13	14
CT weight	108	84	69	108	103	126	129	79	64	130	93	98	78	59
CT weight avg.	15.4	12.0	9.9	15.4	14.7	18.0	18.4	11.3	9.1	18.6	13.3	14.0	11.1	9.8
CT weight adjusted *											59 * (4)			
PT weight				92	90					63	41 * (4)			
PT weight avg.				13.4	12.8					9	10.25			
CT PT distance				22	23					57	34			
CT PT distance avg.				3.14	3.28					8.14	8.50			

Alphabet PT:	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Shift:	8				3			3						3	3	-3	-3	0	0							

Alphabet CT: I H K Q R N O S T

I hope this solution to the EASTNORTHEAST passage helps cracking the remaining of K4, my attempts to untangle blocks #10 & #11 of known plaintext have failed to produce any viable results due to a lack of usable coincidences. A few strong clues indicate blocks #10 & #11 ciphers unlikely to share parameters with blocks #4 & #5, and also not among themselves. Here's a few reasons:

-Using blocks #4 & #5 Layer-2 parameters on block #10 creates a "T" collision on letters #4 & #5, hence block #10 doesn't seem to use the known layer-2 parameters.

-On Block #11 letter #4 both the cipher-text and plaintext is letter "K", this outcome is not possible by using the known cipher parameters from Block #4 & #5.

-Block #10 and #11 position #3 cipher-text is both "P" while the expected plaintext is "R" and "C", this collision wouldn't happen if the blocks shared the same cipher parameters.

-Block #10 plaintext "E" and Block #11 "O" share the same position #2, assuming Layer-2 position shift is the same for both letters, if Layer-1 substitution was the same as seen in blocks #4 & #5 these should show at an equal distance vs their correspondences. In blocks #4 & #5 once Layer-2 is solved, known "E" and "O" are at an equal distance of 3 from their Layer-1 cipher correspondences "H" and "R".

No other K4 blocks seem to follow the cipher/parameter mechanics of blocks #4 & #5. Analysis of blocks #3 and #6 produces favorable Layer2 results but nothing conclusive, no english language is produced when applying the known layer1 parameters.

That's all for now, Eduardo